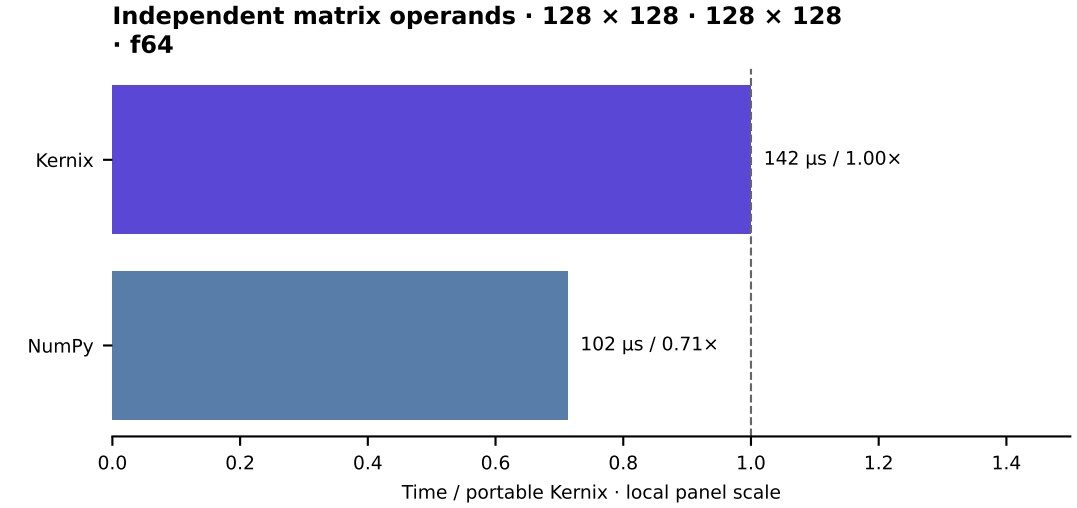
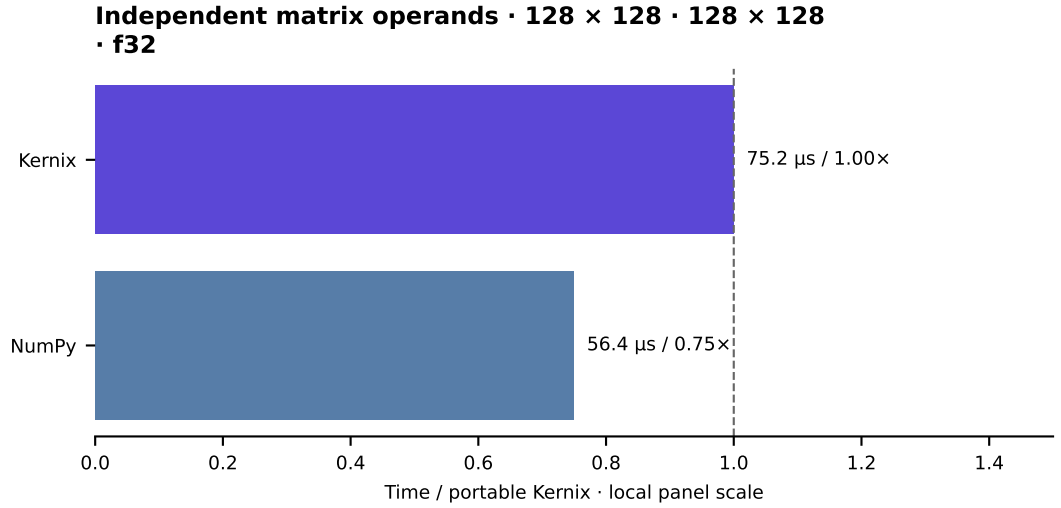
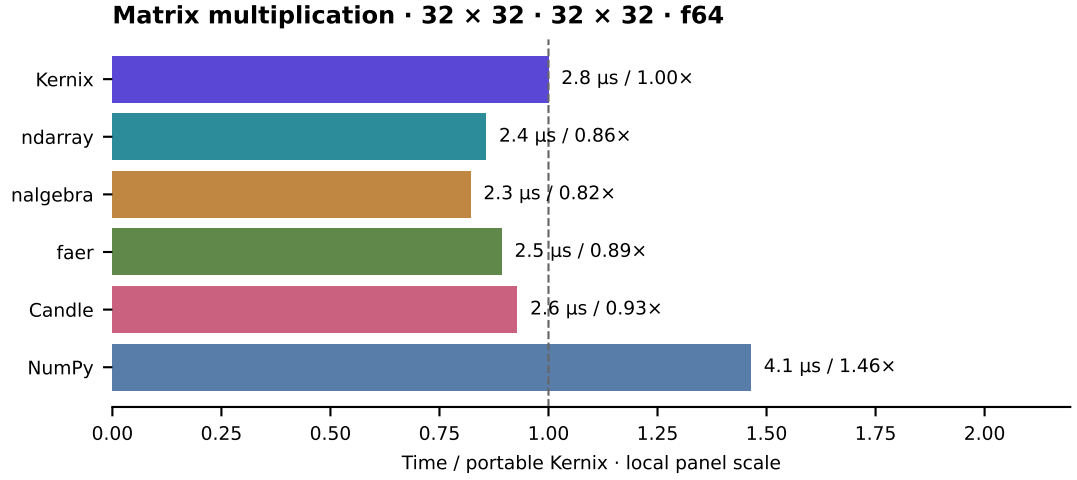
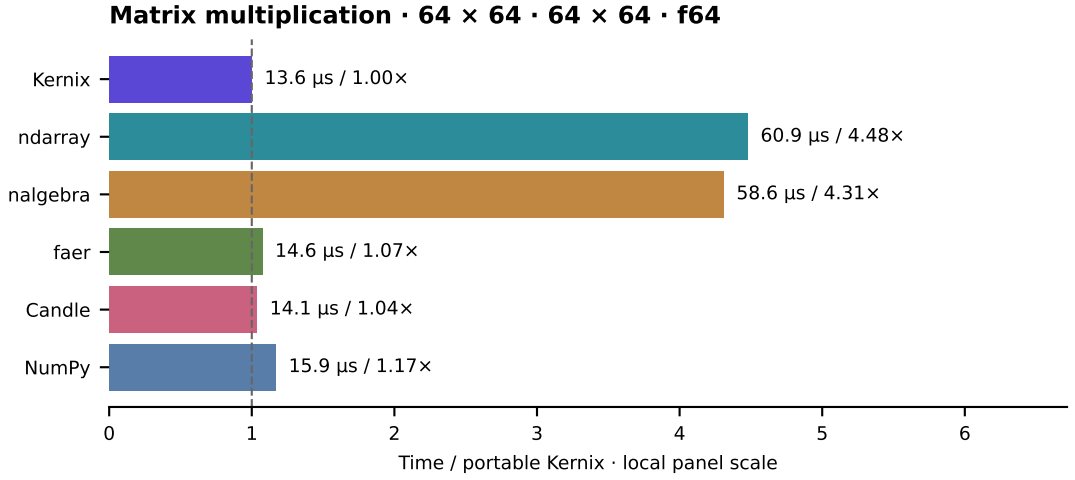
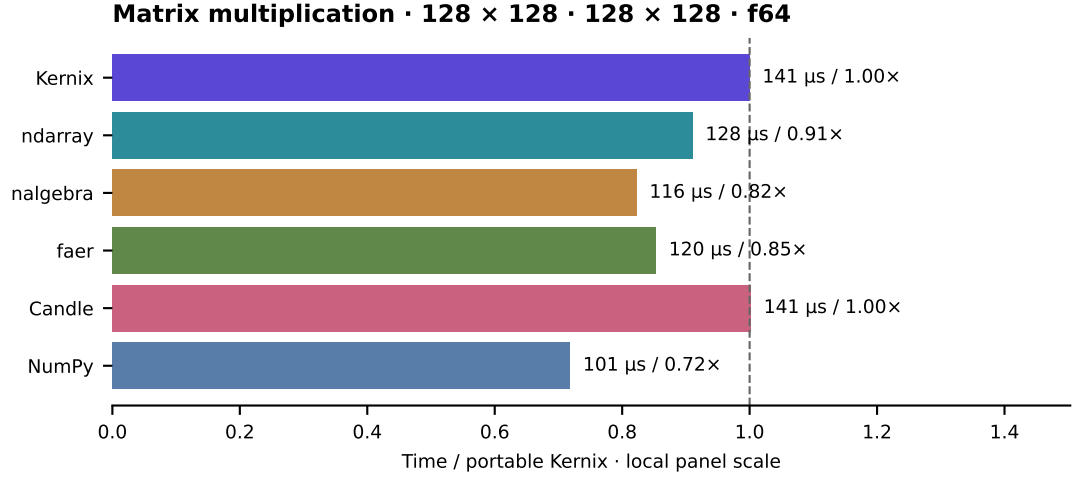
Matrix multiplication · page 1/9

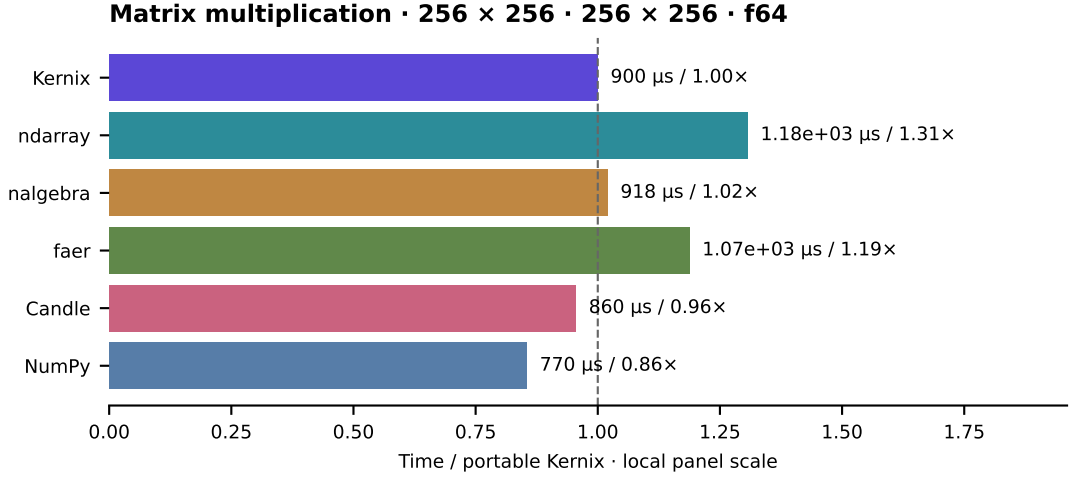




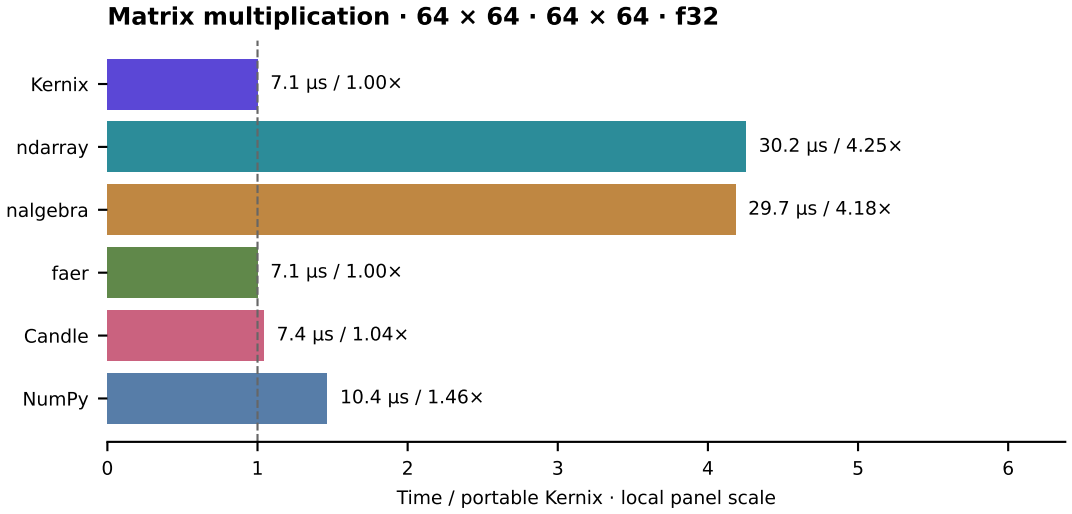


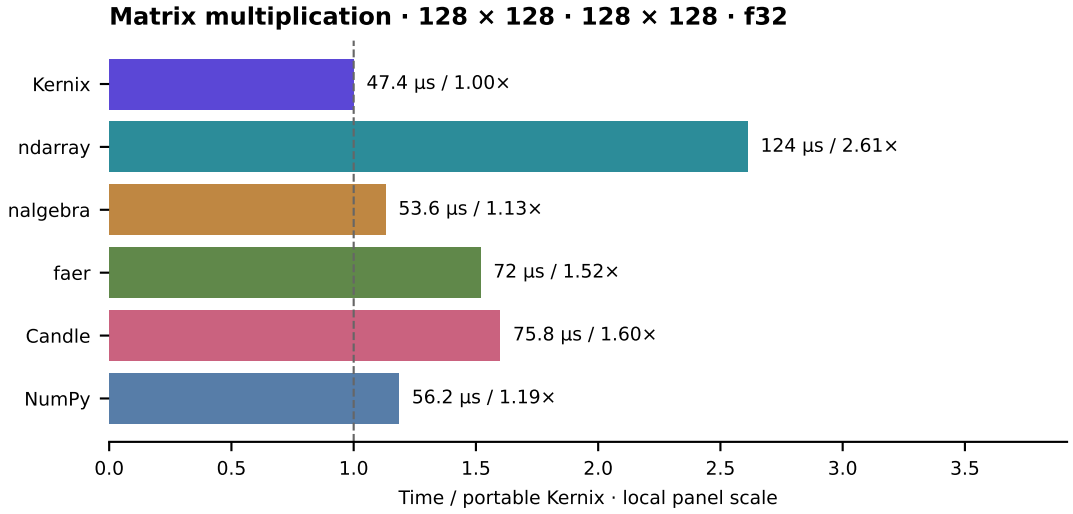


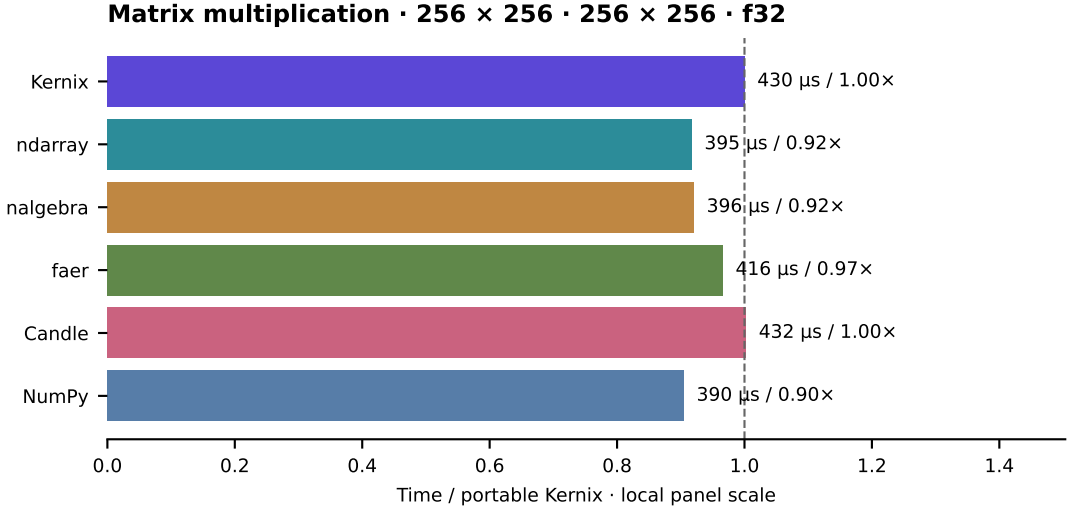


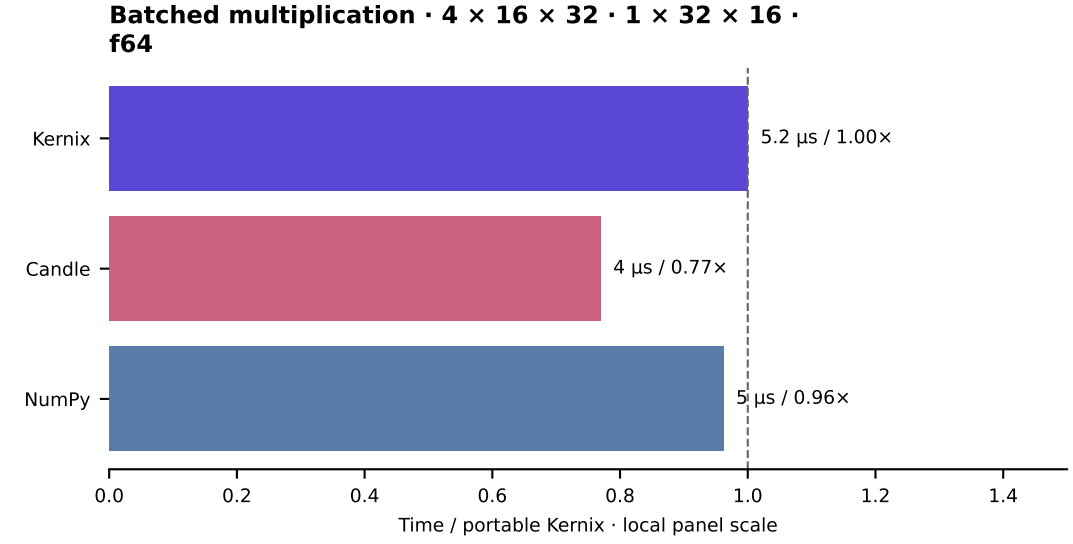


Matrix multiplication · page 2/9









Matmul · 2 workers · reused output · 64 × 64 · 64 × 64 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 256 × 256 · 256 × 256 · f64

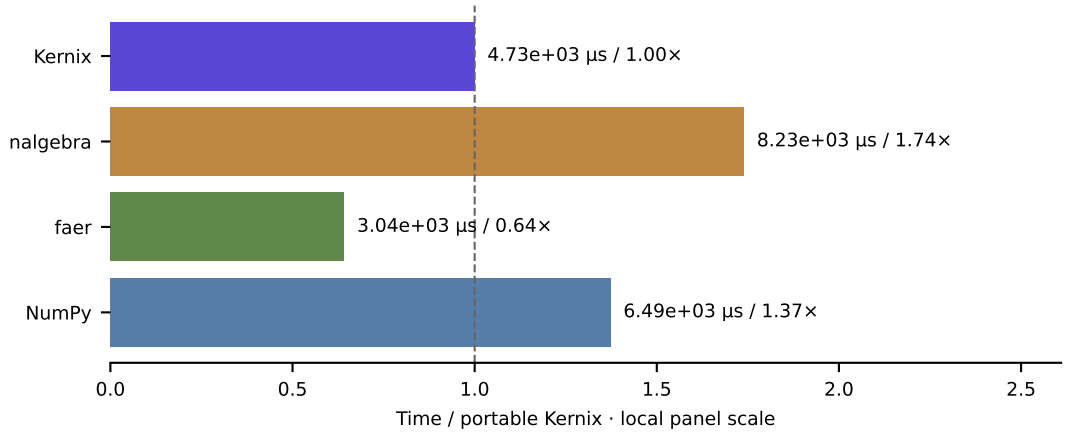
Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f32

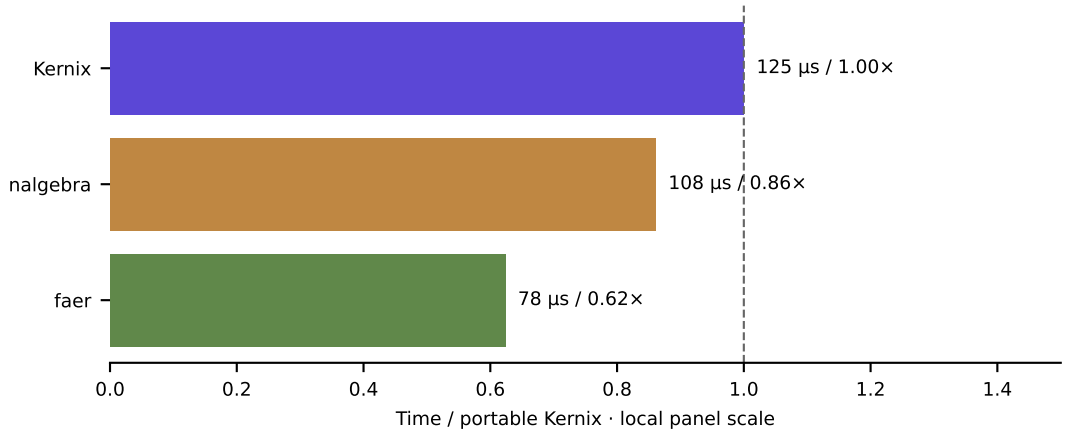
Previous-version comparison only; no peers measured

Decompositions and solvers · page 4/9

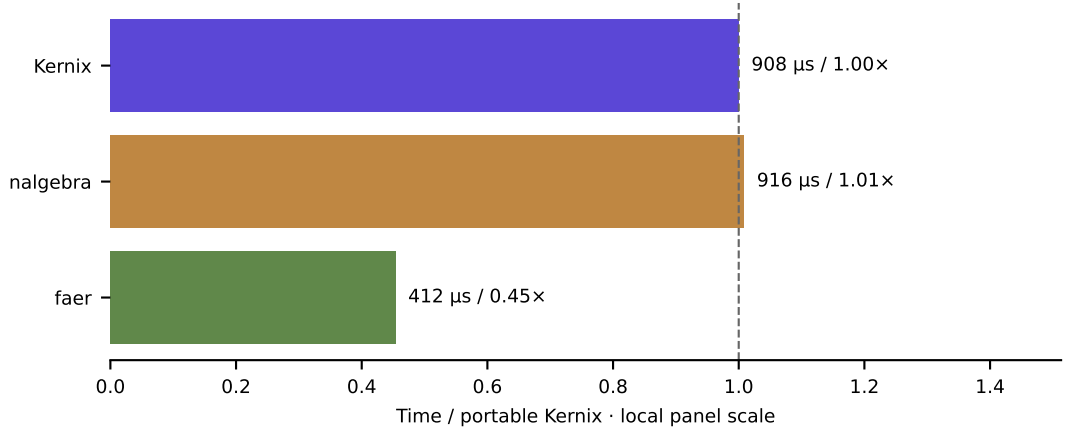
Cholesky factorization · 512 × 512 · f64



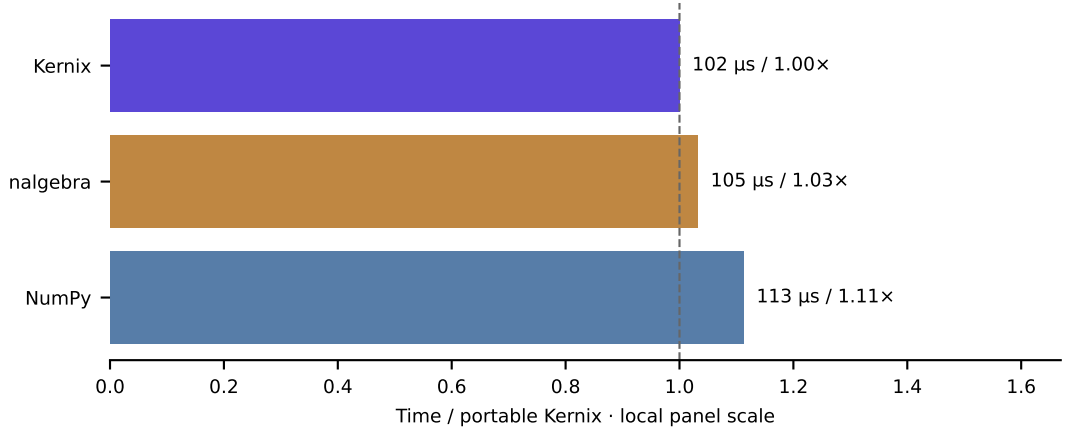
Column-pivoted QR · 96 × 32 · f64



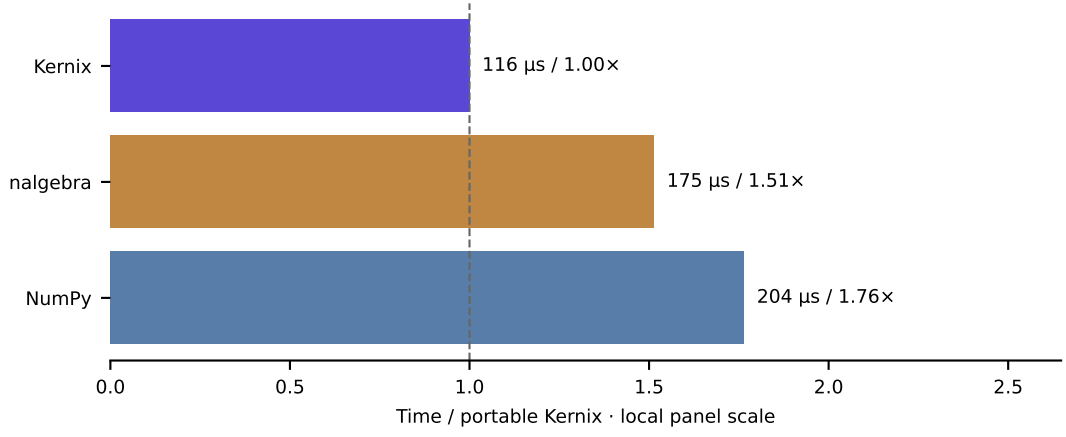
Column-pivoted QR · 256 × 64 · f64



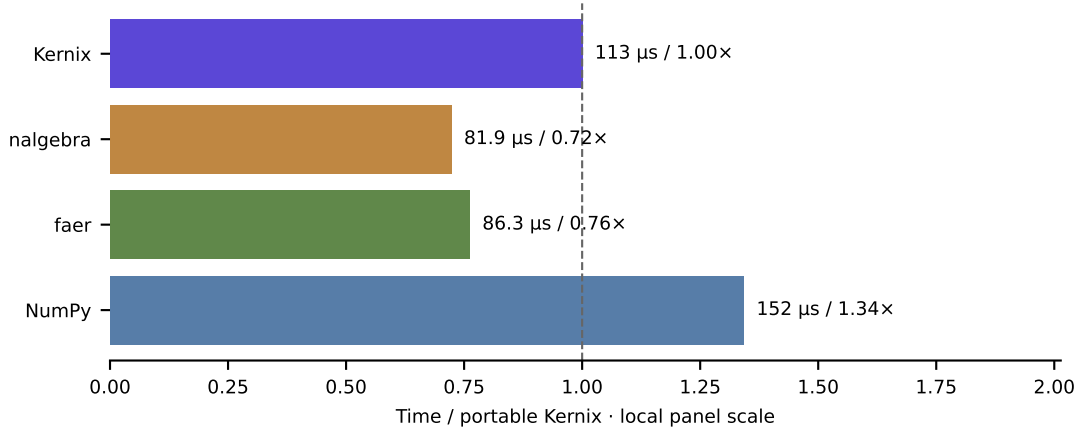
Minimum-norm least squares · 64 × 32, rank 16, 4 RHS · f64



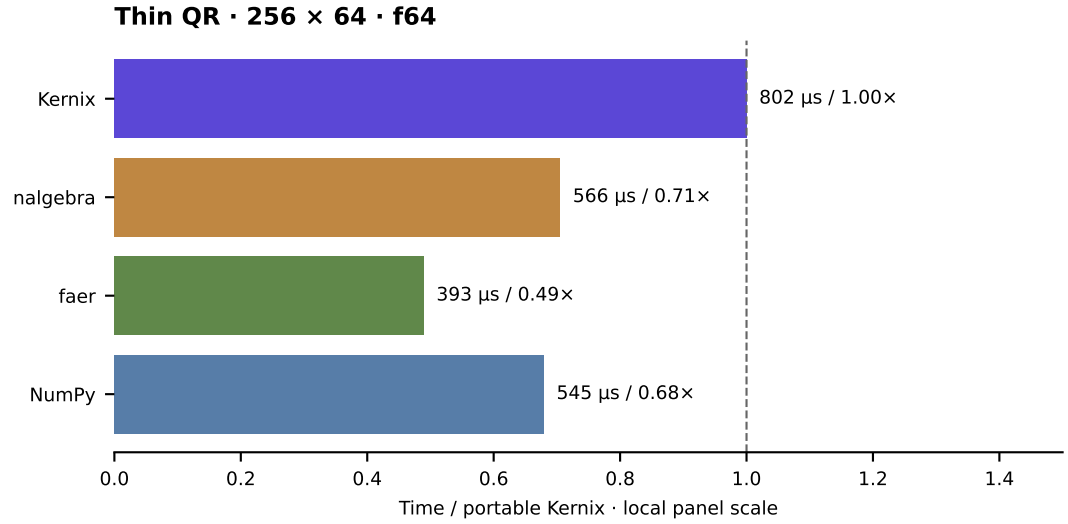
Minimum-norm least squares · 32 × 64, rank 32, 4 RHS · f64

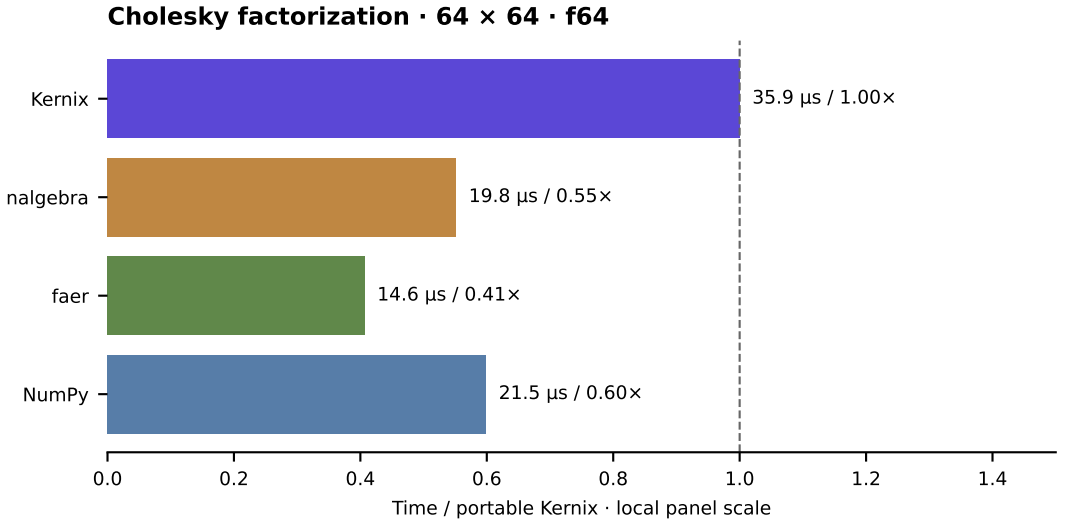


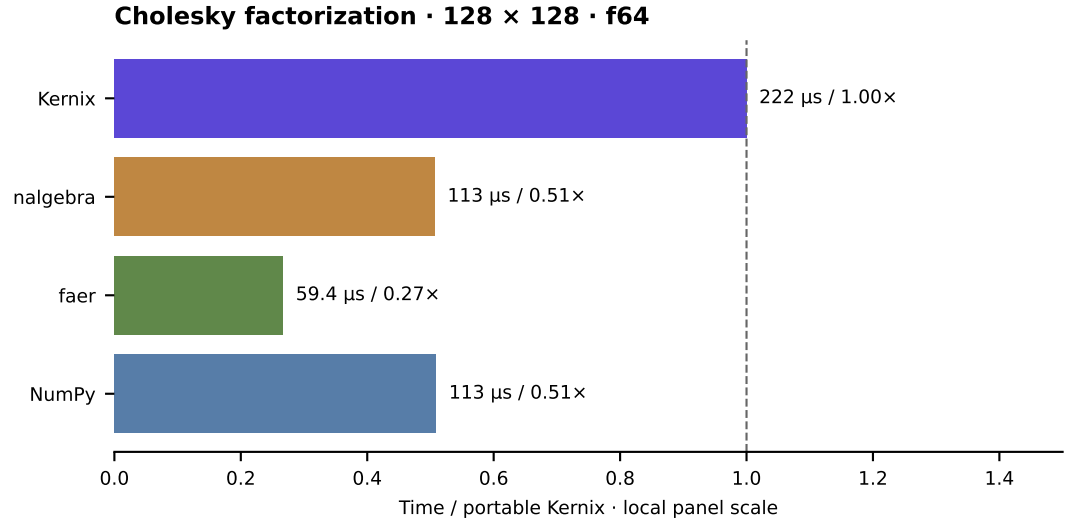
Thin QR · 96 × 32 · f64

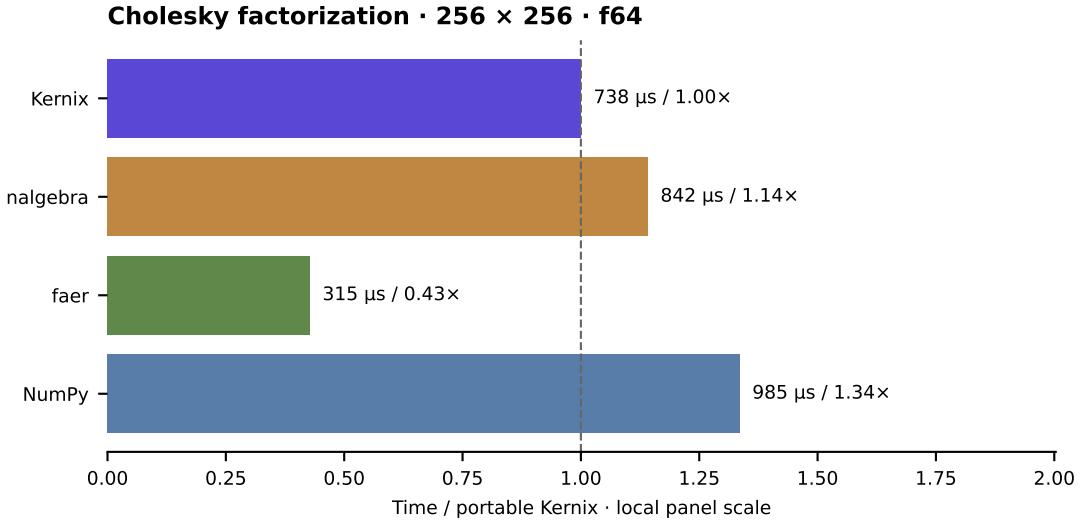


Decompositions and solvers · page 5/9



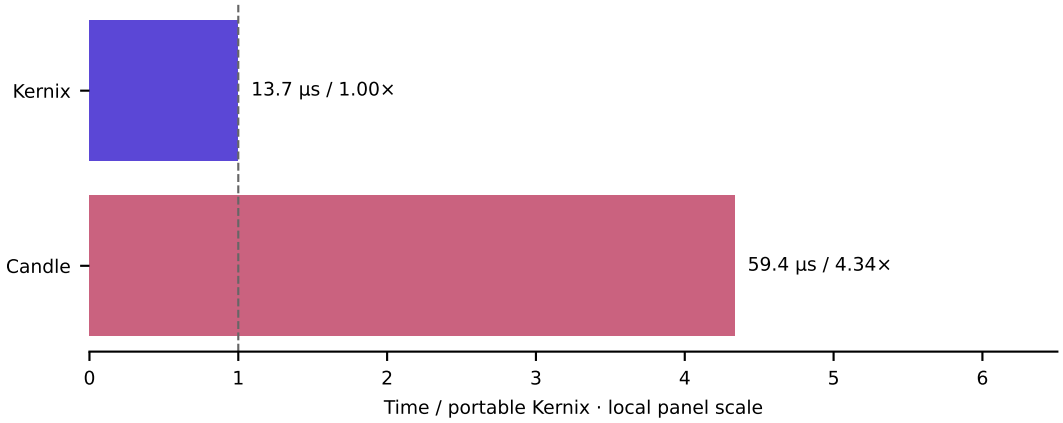




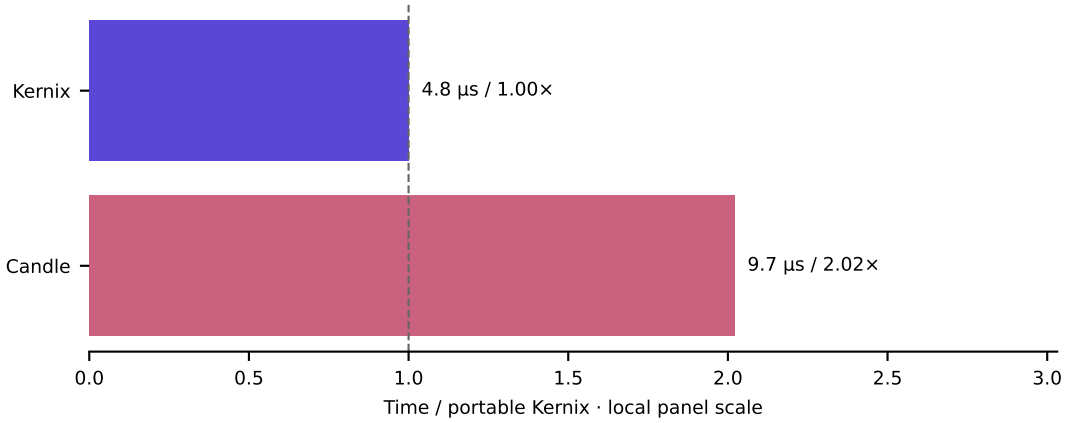


Training and autodiff · page 6/9

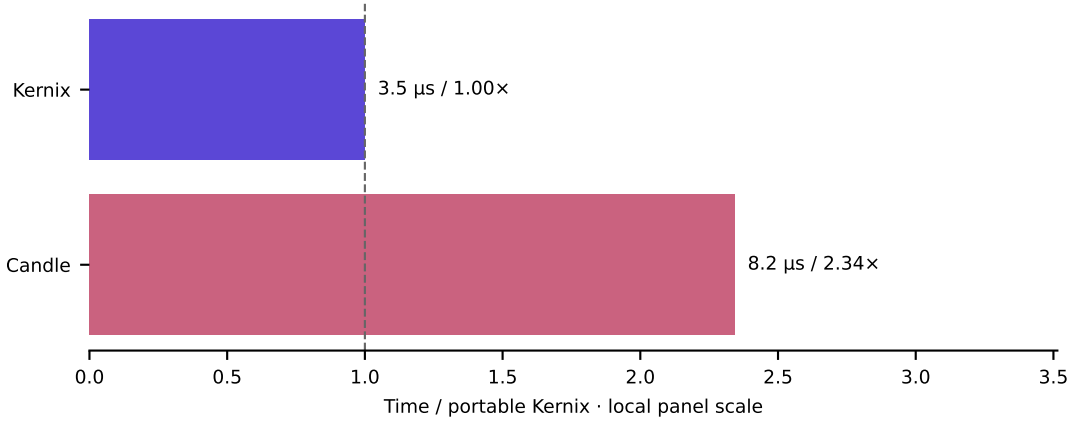
Shared-variable backward · sum(X * X), 64 × 32; prepared graph · f64



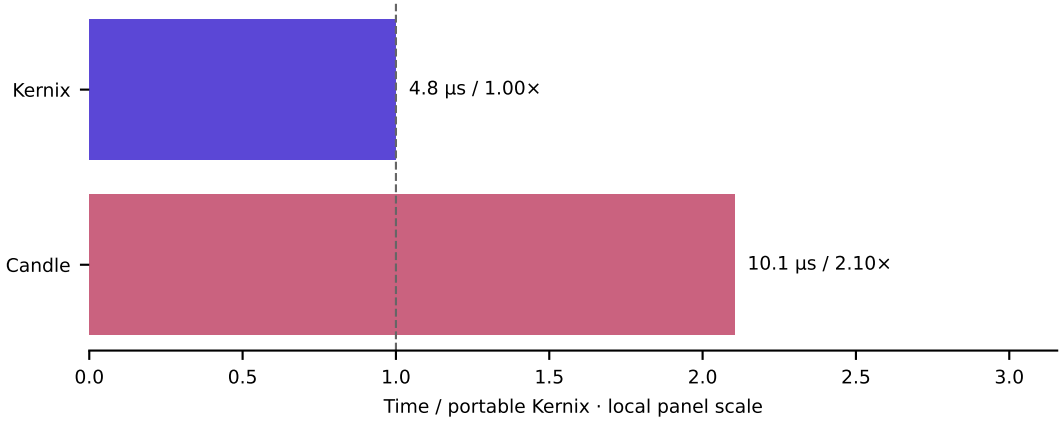
Record + backward · sum(A · W), 32 × 16 · 16 × 8 · f64



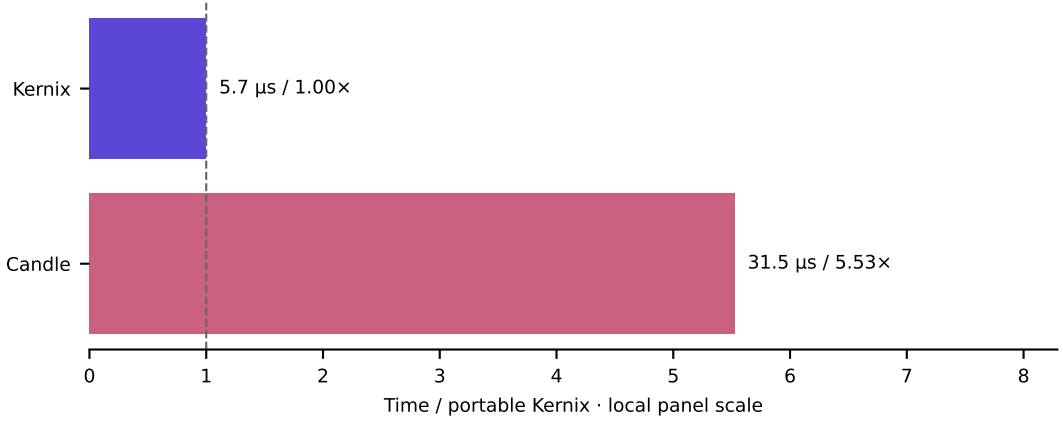
Model inference · batched normalized/sliced MLP · f64



Model recording + forward · batched normalized/sliced MLP · f64



Model backward · prepared graph → fresh gradients · f64

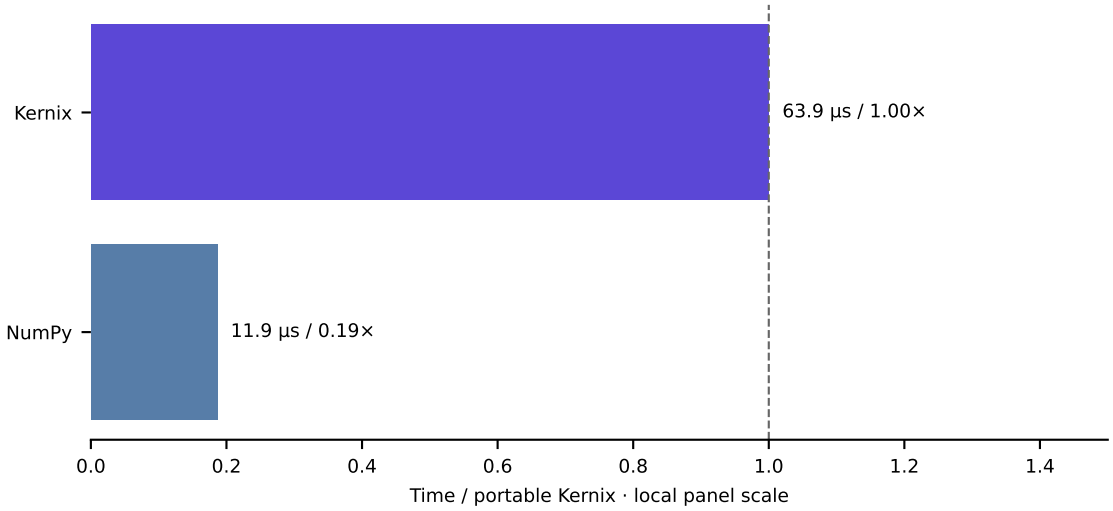


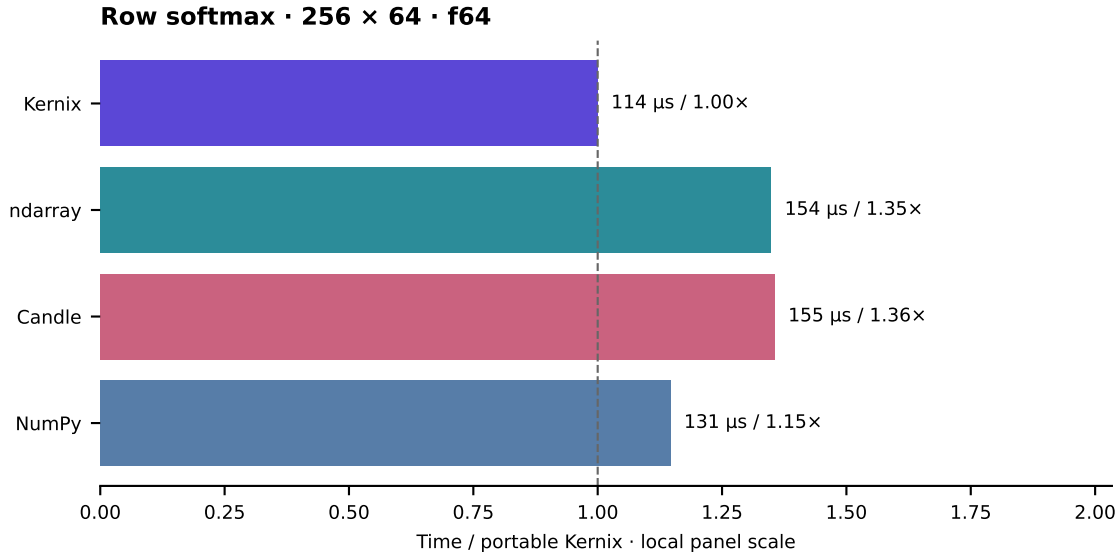
Parameter update · transactional SGD, fixed gradients · f64

Previous-version comparison only; no peers measured

Tensor contractions · page 7/9

Coalesced tensor contraction · 16 × 32 × 8 · 32 × 8 × 16 · f64




Row softmax · 256 × 64 · f64
Kernix
ndarray
Candle
NumPy
114 µs / 1.00×
154 µs / 1.35×
155 µs / 1.36×
131 µs / 1.15×
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time / portable Kernix · local panel scale

Tensor construction · page 9/9

